.3 Pseudocode

Surface BlastDesign Calculator

```
BD = INPUT ("Enter Blasting Diameter (BD)")
ρ = INPUT ("Enter Average In-Hole Density (ρ)")
a = INPUT ("Enter Spacing to Burden ratio (a)")
BH = INPUT ("Enter Bench Height (BH)")
SD = INPUT ("Enter Sub-drill (SD)")
PFtech = INPUT ("Enter Technical Powder Factor (PFtech)")
Mc = (ρ * BD**2) / 1273
SL = 20 * (BD / 1000)
IF (a * PFtech * BH) == 0 THEN
        RETURN ERROR("Division by zero - check inputs a, PFtech, BH")
        END IF
B = sqrt ( (Mc * (BH - SL - SD)) / (a * PFtech * BH) )
S = a * B spacing, derived from S/B ratio
H = BH  (assumed same as bench height)
IF (B * S * H) == 0 THEN
        RETURN ERROR("Division by zero - check B, S, H")
        END IF
B = sqrt ( (Mc * (BH - SL - SD)) / (a * PFtech * BH) )
S = a * B (spacing, derived from S/B ratio)
H = BH
IF (B * S * H) == 0 THEN
        RETURN ERROR("Division by zero - check B, S, H")
        END IF
ActualPowderFactor = ((BH - SL + SD) * Mc) / (B * S * H)
OUTPUT("Stemming Length (SL) = ", SL)
OUTPUT("Burden (B) = ", B)
OUTPUT("Actual Powder Factor = ", ActualPowderFactor)
END FUNCTION
```

Fragmentation Calculator

```
BH = INPUT ("Enter Bench Height (BH)")
ρ = INPUT ("Enter Average In-Hole Density (ρ)")
E = INPUT ("Enter Relative Energy of Explosive (E)")
SL = INPUT ("Enter Stemming Length (SL)")
A = INPUT ("Enter Rock Factor (A)")
K = INPUT ("Enter Powder Factor (K)")
Q = INPUT ("Enter Mass of Explosive per Hole (Q)")
X50 = A * (K **-0.8) * (Q **(1/6)) * ((115 / E) **(19/20))
OUTPUT("Mean Fragmentation Size (X50) = ", X50)
END FUNCTION
```